

• LIVE ON KITE AI

UnlockShield

An autonomous portfolio risk agent.

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Submitted to the Kite AI Global Hackathon — Track 2: Agentic Trading & Portfolio Management

Token unlocks are public schedules that destroy DeFi positions on cue.

OCT 30, 2024

TIA

16.3% of supply unlocked

−28.5%

7-day price impact

JUL 24, 2024

WLD

6.7% of supply unlocked

−22.8%

7-day price impact

AUG 15, 2024

SEI

1.38% of supply unlocked

−18.9%

7-day price impact

Existing tools tell you when unlocks happen. None forecast the magnitude, simulate the damage, or prove their predictions weren't cherry-picked after the fact.

A risk agent that actually trades — bounded by code, settled in USDC.

UnlockShield is a fully autonomous portfolio risk agent. Every 90 seconds it scans 38 tokens, scores each on a 12-factor risk model, runs regime-switching Monte Carlo, commits the forecast on Kite, and moves USDC through an on-chain spending policy. No human in the loop.

01

Autonomous

90-second cycles run continuously. The agent scans, scores, commits and settles without user input.

02

Bounded

Spending policy enforced in Solidity. The agent's key cannot widen its own bounds — only the owner can.

03

Verifiable

Every prediction is a keccak256 commit on Kite. Every hedge is a USDC transfer with a public tx hash.

Six stages. Ninety seconds. No clicks.

1 Monitor LIVE DATA

Ingest CoinPaprika, DeFiLlama, Tokenomist, F&G, Polymarket.

2 Score 12-FACTOR

Composite risk per token from 12 weighted signals.

3 Simulate RS-GARCH

2,000 Monte Carlo paths with jump-diffusion stress.

4 Commit KITE ORACLE

keccak256 hash committed on Kite before the event.

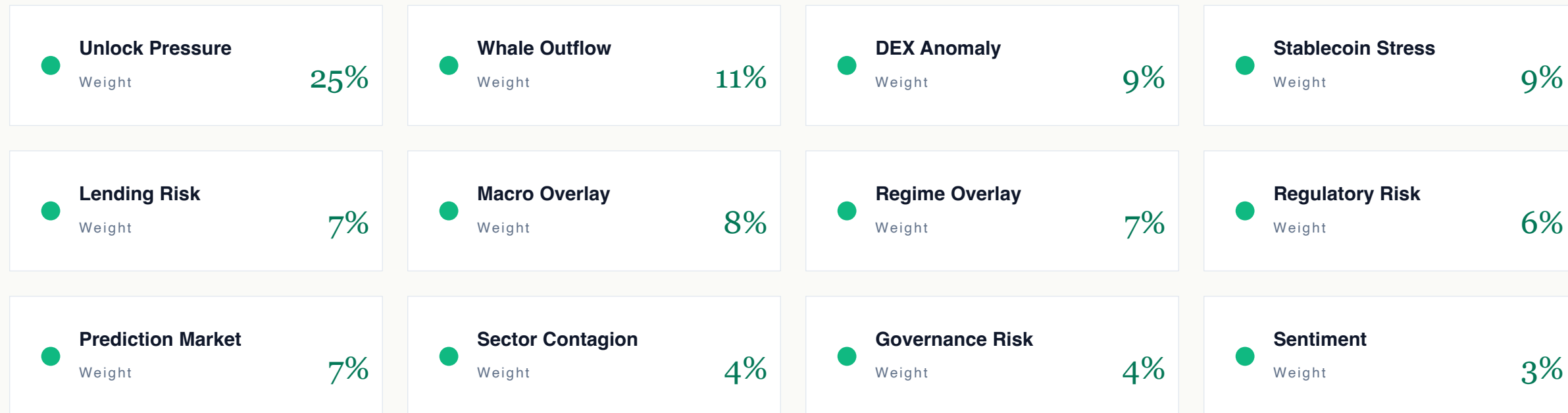
5 Execute USDC POLICY

AgentTreasury.executeHedge() transfers USDC per gate.

6 Verify REPUTATION

After event, reveal + score → on-chain reputation grade.

Every token, every cycle — scored on twelve independent signals.



Each signal returns a 0–100 score with explicit weight and contribution. The composite is a transparent weighted sum — every factor is independently auditable in the activity feed.

Regime-Switching GARCH + Merton Jump-Diffusion. Two thousand paths.

2,000

Monte Carlo paths per simulation

180+

Historical unlock events calibrating the model

3

Markov regime states (BULL / BEAR / SIDEWAYS)

969

Lines of Python in stress_engine.py

ACADEMIC FOUNDATION

Bollerslev (1986)

GARCH(1,1) — volatility clustering

Merton (1976)

Jump-diffusion for sudden shocks

Hamilton (1989)

Markov regime-switching

Milionis (2022)

Loss-Versus-Rebalancing for LPs

Three contracts. One chain. Everything verifiable.

Kite AI Testnet · Chain ID 2368 · `rpc-testnet.gokite.ai`

UnlockShieldOracle

`0xD2d642Ea44973d90Bb0a6f403e8A4815020Fdd79`

Commit-reveal prediction registry. keccak256 hashes committed before each unlock event, revealed and scored after.

AgentTreasury

`0xf6D48bDACadA5105808d04B3719605B3AfbAf9Bb`

Bounded-autonomy capital execution. Holds USDC, enforces 4 on-chain spending gates per executeHedge call.

MockUSDC

`0xe48084773d0e60b6272008a2F87B36Ccf1d8eB02`

6-decimal stablecoin for testnet settlement. Every hedge is a real USDC transfer; native KITE only used for gas.

Spending policy written in Solidity. The agent cannot widen its own bounds.

MIN RISK THRESHOLD

≥ 25

Composite score required to act

MAX SINGLE TRADE

\$1,000

Capped per executeHedge call

DAILY CAP (ROLLING 24H)

\$5,000

Resets on a 24-hour bucket

TREASURY BALANCE

Required

Trade only if USDC available

AgentTreasury.sol

```
if (riskScore < minRiskScore)
    → HedgeBlocked

if (amountUsd > maxSingleTrade)
    → HedgeBlocked

if (spentToday + amount
    > dailyCapUsd)
    → HedgeBlocked

usdc.transfer(recipient,
              amount)
emit HedgeExecuted(...)
```


Real trades. Real USDC. Real reputation grade.

74

Trades on-chain

Each is a real USDC transfer

\$7.9K

USDC deployed

From a self-funded treasury

C+

Reputation grade

Honest score from 13 backtests

597+

Hedges policy-blocked

Bounded autonomy working as designed

BACKTESTED ON 13 REAL 2024-2025 UNLOCK EVENTS

ARB · OP · APT · TIA · SUI · SEI

\$6,967 saved · 61.5% win rate · TIA -28.5% covered

Same scoring formula runs on every future prediction. No tuning to inflate the grade.

Every requirement, with proof.

✓ Autonomous Agent

agent_loop.py runs every 90s, no user input. Verified on /api/agent/activity.

✓ Executes Paid Actions

AgentTreasury.executeHedge() transfers USDC. 74 trades on-chain.

✓ Settles on Kite

All commits + hedges land on Kite Testnet, Chain ID 2368.

✓ Stablecoin-First

Every capital action moves USDC; KITE only used for gas.

✓ Programmable Constraints

Min risk, max trade, daily cap — enforced inside the treasury contract.

✓ Reputation-Aware

On-chain grade (S-F, 0-1000) from commit-reveal scored predictions.

Citadel-style multi-strategy risk infrastructure — settled in USDC, audited on Kite.